

Spatial Differences in Copper and Lead Levels in Seagrasses and Seaweeds from Curaçaoan Bays

Authors: Valenzuela, Emily, Bilimoria, Aliya, Ruggeri, Ellianna, Vitales, Isabella, Penesso, Giorgia, et al.

Source: Caribbean Journal of Science, 56(1) : 207-223

Published By: University of Puerto Rico at Mayagüez

URL: <https://doi.org/10.18475/cjos.v56i1.a18>

The BioOne Digital Library (<https://bioone.org/>) provides worldwide distribution for more than 580 journals and eBooks from BioOne's community of over 150 nonprofit societies, research institutions, and university presses in the biological, ecological, and environmental sciences. The BioOne Digital Library encompasses the flagship aggregation BioOne Complete (<https://bioone.org/subscribe>), the BioOne Complete Archive (<https://bioone.org/archive>), and the BioOne eBooks program offerings ESA eBook Collection (<https://bioone.org/esa-ebooks>) and CSIRO Publishing BioSelect Collection (<https://bioone.org/csiro-ebooks>).

Your use of this PDF, the BioOne Digital Library, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Digital Library content is strictly limited to personal, educational, and non-commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne is an innovative nonprofit that sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Spatial Differences in Copper and Lead Levels in Seagrasses and Seaweeds from Curaçaoan Bays

EMILY VALENZUELA¹, ALIYA BILIMORIA¹, ELLIANNA RUGGERI¹, ISABELLA VITALES¹, GIORGIA PENESSO¹, JUSTIN O'PELLA², INGO VAN VEGHEL³, ARD VREUGDENHIL³, DAVIS M. ASHLEY¹, MANUELA TRIPEPI¹, AND JEFFREY T. F. ASHLEY^{1,*}

¹*Department of Biological and Chemical Sciences, College of Life Sciences, Thomas Jefferson University, Philadelphia, Pennsylvania, U. S. A.*

²*Kanbar College of Design, Engineering and Commerce, Thomas Jefferson University, Philadelphia, Pennsylvania, U. S. A.*

³*Sea Turtle Conservation Curaçao, Curaçao*

**Corresponding author: Jeffrey.ashley@jefferson.edu*

ABSTRACT—The health of aquatic coastal ecosystems in the Caribbean Sea is vital for retaining their ecological integrity and maintaining the economic value as desirable destinations to visit. A suite of contaminants, including heavy metals, may be detrimental to these ecosystems' viability, yet datasets and subsequent environmental assessments are somewhat minimal. To that end, this study used various species of seagrasses and seaweeds as bioindicators of copper and lead levels at five adjacent land sites of varying use on Curaçao. Small but statistically significant differences in metal concentrations were observed between the five sites and were largely driven by differences in site location, not plant species. Awa di Oospunt, the most remote and less human-impacted site, had mean ($\pm$ standard deviation) copper and lead concentrations of 2.38 ($\pm$ 1.46) and 0.60 ($\pm$ 0.18) ng/g or ppb dry weight (dw), respectively. Conversely, seagrasses from Caracas Bay, adjacent to more heavily urbanized and industrialized land uses, had mean copper and lead concentrations of 5.86 ($\pm$ 2.03) and 2.83 ($\pm$ 0.61) ng/g dw, respectively. Differences in adjacent land-use and proximity to wastewater and industrial discharges were likely primary factors in determining observed concentrations. No site harbored elevated levels of concern based on comparison to other studies. Nonetheless, organisms feeding upon seagrasses, such as green sea turtles, may be exposed to these metals through ingestion. Future studies should expand on the spatial and temporal concentration differences with a broader suite of potential contaminants, evaluate use of different marine species as bioindicators, and more definitively identify land-based sources and inputs of potential contaminants.

For decades, the waters, sediment, and biota of the Caribbean Sea have been impacted by pollution from a suite of anthropogenic activities (Ross and DeLorenzo 1997). In an early review article summarizing known contaminants (e.g., petrochemical hydrocarbons, marine debris, industrial waste, and sewage/nutrients) in the Caribbean Sea, Siung-Chang (1997) concluded that marine pollution was becoming a significant problem in the region, especially in coastal environments. In the decades since this article, researchers have garnered more insight into the types of pollutants, their sources, and their transport mechanisms, promoting more urgent calls for measures to assess and limit pollution within the region (Diez 2019; Guyvenchy et al. 2023).

Trace metals, especially those considered to be

potentially toxic, are one class of pollutants that may reduce the integrity of marine systems. To date, evaluating the magnitude, spatial extent of contamination, and the sources of trace metals within biotic (flora and fauna) and abiotic (water, sediment, porewater) components of marine ecosystems of the Caribbean Sea has been limited. For example, reviewing literature on anthropogenic trace metal and organic compounds in the Wider Caribbean Region (WCR), Fernandez et al. (2007) noted that most studies centered on the South and Central American Caribbean region, with little to no information from most nations in the WCR, especially small island countries. More recently, Hardy et al. (2025) reviewed potentially toxic element data published between 1990 and 2020, concluding that levels

in sediments and aquatic samples from various locations indicated possible environmental, toxicological, and public health risks, and pointed to the need for more regional-scale assessments.

Trace metal sources may be natural (e.g., rock weathering, volcanic eruptions, wind-borne particles) or anthropogenic (e.g., domestic, industrial, and agricultural processes, fossil fuel combustion, incineration of waste). Trace metals may enter coastal marine environments through mechanisms such as direct discharges of wastewaters (Sanz et al. 2021), urban and industrial runoff (Joshi and Balasubramanian 2010), wet and dry atmospheric deposition (Duce et al. 1991; Omrani et al. 2017), aeolian transport of mining dust (Kasongo et al. 2024), and influx via currents (Bertoncelj et al. 2025). Once in the aquatic environment, metal speciation, known as the chemical form which the element is present, may vary depending on factors such as the metal itself, and physicochemical properties of the aquatic environment (e.g., salinity, reduction/oxidation potential, and dissolved organic carbon). Speciation includes all potential forms of that element from dissolved metal ions (termed 'free ion') or dissolved complexes, to colloidal or particle-bound forms (Al Naggat et al. 2018). The speciation of a metal determines the degree and mode of transport within a coastal system and its bioavailability. For most trace metals, the free ion is the form most available to both marine flora and fauna.

Trace metals may also be classified as either essential (e.g., copper, iron, zinc) or non-essential (e.g., lead, cadmium, arsenic, mercury), based on whether or not they are critical to biological functioning. Important determinants of ecotoxicological effects are the concentration of both essential and non-essential trace metals, the exposed organism, and the exposure time, among other factors. Trace metals may pose a suite of deleterious effects to marine organisms, exacerbated by their non-biodegradable, durable nature (Shahjahan et al. 2022). Trace metal accumulation in marine primary producers represents the first step in the potential for trophic transfer to higher organisms. Reproduction, behavior, and cellular abnormalities have been found to be a potential result of metal accumulation from an organism's diet (Kolarova and Napiorkowski 2021).

Assessing the magnitude and extent of trace metal contamination in marine systems requires time and effort, both to collect appropriate numbers of abiotic

and biotic samples and to perform subsequent chemical analysis of them. Over the past decades, researchers have used marine floral and faunal species as sentinels or indicators for environmental pollution and ecosystem health (Savoca et al. 2025). Parmar et al. (2016) describe bioindicators as the use of taxa to detect the presence of pollution due to the species' resistance to ecological variation. Faunal species, exposed passively to dissolved metal species through respiration and dietary ingestion of contaminated prey, have been used as bioindicators, acting as surrogates to assess trace metal exposure and accumulation in other marine species or humans. Equally, bioindicators can shed light on trace metal sources, transport, and fate processes in coastal ecosystems. For example, sea cucumbers exposed to contaminated coastal sediments in Cispata Bay, Columbia, were used to identify agricultural activities as primary mechanism for delivering trace metals to that ecosystem (Marrugo-Negret et al. 2021). Likewise, more sessile organisms like filter-feeding bivalves have been successfully used as bioindicators of trace metal pollution. For example, in a temporal study over several years, Ostapczuk et al. (1997) used mussels, as well as alga, and concluded that trace metal accumulation in these species was suitable for demonstrating long-term concentration changes in marine coastal ecosystems.

For decades, seagrass/seaweeds have been a popular choice as bioindicators due to their ubiquitous presence and non-motility, which support their use as proxies for local/regional pollutant inventories (Martínez-Crego et al. 2008; Govers et al. 2014; Baptista et al. 2024). Seagrasses are abundant in shallow waters and cover over 600,000 km² of the coastal oceans (Unsworth and Cullen-Unsworth 2017). They are a common source of food for many aquatic species, such as fish and sea turtles, and act as protective environments from predators (Unsworth and Cullen-Unsworth 2017). However, seagrass beds may be characterized as threatened ecosystems due to direct mechanical destruction from trawling and anchoring, and through pollution (Schalacher-Hoenlinger and Schalacher 1998; Romeo et al. 1995). For example, Govers et al. (2014) observed negative effects of eutrophication in Curaçaoan seagrass beds adjacent to outflows of waste waters and coastal housing developments. Blades/leaves of seagrass have been proven to accumulate heavy metals, either directly from the water or via root transfer from metals in sed-

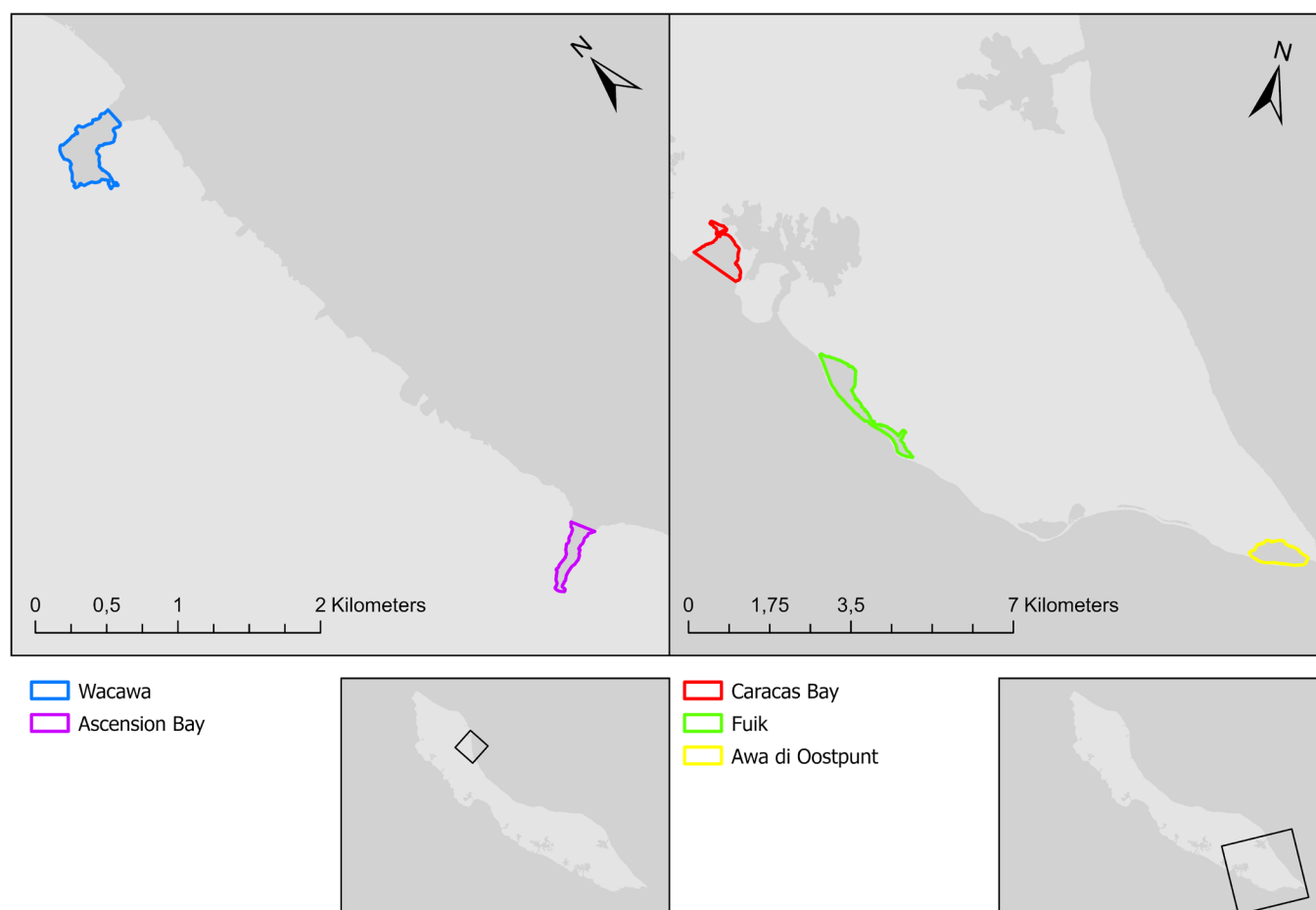


FIGURE 1. Areas of seagrass and seaweed collection at Wacawa, Ascension Bay, Caracas Bay, Fuik, and Awa di Oostpunt on Curaçao.

iment (Tupan and Azriagishih 2016; Jeong et al. 2021; Wang et al. 2025). Depending on the species of seagrass, different trace metals may accumulate to varying degrees (Mishra et al. 2022). Elevated levels of trace metals in seagrasses may negatively affect an ecosystem's health, both for the seagrasses and for species that feed upon them.

To address the need for more trace metal data from understudied coastal areas within the Caribbean Sea, this study focused on coastal aquatic ecosystems surrounding the island of Curaçao. An island in the south-western Caribbean, Curaçao includes plentiful coral reefs, mangroves, and terrestrial biodiversity, though increasing human developments have consequently impacted the natural aspects of the island. Like other picturesque island nations in the WCR, Curaçao has seen increasing tourists and population in the past decades; these have prompted increasing coastal development, and subsequent increases in municipal and industrial discharges, making Curaçao a suitable sampling site for investigating potential release of heavy metals.

In this study, numerous species of seagrasses/seaweeds were collected from various Curaçaoan bays adjacent to differing land use with the aim of assessing the concentrations and identifying potential sources and inputs of trace metals to the coastal ecosystems. We focused on two metals, copper and lead, that exhibit different aquatic chemistries and sources.

METHODS AND MATERIALS

Study Area and Sampling Sites

Curaçao, consisting of the 444 km² main island and the much smaller uninhabited island of Klein Curaçao, lies about 65 km north of Venezuela in the southern Caribbean Sea. The island has a population of 155, 826 (2023) with most living in or around the capital of Willemstad, covering about one third of the island. Five sites, two located in the northeast of the island (Wacawa and Ascencion), and three in the southwest (Caracas Bay, Fuik Bay, and Awa di Oostpunt) (Fig. 1), were chosen for this study, representing bays or enclaves adjacent to differing land uses.

Caracas Bay is a bay adjacent to the southeastern part of Jan Thiel, a neighborhood in southeastern Willemstad (Fig. 2). The bay lies between Jan Thiel Bay to the west and Spanish Water to the east. Geologically, Caracas Bay was the site of a carbonate terrace slope failure, also known as the Caracas Bay Slide, which caused a carbonate cuesta to fall free to the ocean (Hornbach et al. 2010). Presently, the bay is adjacent to one of the most highly developed and urbanized areas on the island. Caracas Bay was a former oil terminal that ceased operation in 1996, though its metal structures still stand. The inner bay has a marina, subsequently boat traffic is high in this area.

Fuik Bay is located on the southwest coast of Curaçao, the southeast neighbor of Caracas Bay. The bay is an elongated, narrow inner bay that runs parallel to the coastline and is in open communication with the sea. The channel has a width of 44.8 meters and a depth of 7.92 meters (Fig. 2). Lying on the hillside of the bay and clearly visible in aerial images (Fig. 2), the Curaçao Mining Company operates a mine that, by quarrying, crushing, and screening, produces limestone aggregates and sand for export. The company operates two berths for ships to load and discharge within the protected bay.

Awa di Oostpunt is the last bay on the far southwest coast of Curaçao (Fig. 2). The bay is oval-shaped with a width of approximately 500 meters. The bay is only reachable by boat. The shallow sides of the bay are covered with different types of seagrasses and seaweeds, with the depth gradually getting deeper away from the shoreline. The center of the bay consists of a sand pit about 15 meters deep. This represents this study's most remote and isolated site, with no known anthropogenic discharges from land-based activities.

Wacawa Bay is adjacent to Wacawa, a former plantation that is currently used as a military training and shooting range located in the west of the island of Curaçao. The geology of Wacawa consists of the Curaçao Lava Formation, Knip Group, and limestone terraces. Due to the variation in geology, soil, and altitude, the terrain has a varied landscape with a wide variety of vegetation types (Restrepo et al. 2022; Janssen et al. 2023). The limestone coastal area of Wacawa has two inlets, Boka Grandi (south side) and Boca Bartol (north side). The samples were collected from Boka Grandi.

Close to Wacawa, Ascension Bay is an inlet on the northeast side of the island (Fig. 2). The inlet is wide on the seaward side and gets narrower into the inlet. The depth of the water varies from 0.3 meters to several meters deep. Ascension Bay was found to be a popular feeding ground for turtles, though its inner bay was characterized to be heavily polluted (waste dumping) by Govers et al. (2014).

Sample Collection

Both leaves and stems from various species of seagrasses and seaweed (Table 1) were hand collected by snorkeling and diving during the summer months of 2024. Bottom sediment at all sites was sand. The number of individual samples collected at Ascension, Awa di Oostpunt, Caracas Bay, Fuik Bay, and Wacawa are listed in Table 1 with a total number of individual samples collected and ultimately analyzed was 38. Following collection, seagrass and seaweed samples were thoroughly rinsed with tap water to remove sediment, epiphytic algae, and saltwater residue. Samples were dried overnight at 50° C. Dried samples were individually placed in clean plastic bags, labelled, and frozen at -18° C until sample shipment and preparation at the East Falls campus of Thomas Jefferson University (Philadelphia, Pennsylvania, U. S. A.).

Sample Preparation

Approximately 0.2 to 0.4 g of each dried and homogenized sea grass sample were weighed and recorded, then placed into individual acid-cleaned, 50 mL CEM® Xpress Teflon vessels (CEM Corporation, Matthews, North Carolina, U. S. A.). Exactly 10.0 mL of ultra-trace 67-70% (~15M) nitric acid (HNO₃) (Fisher Chemical) was added to each sample (Hassan et al. 2007; Gonzalez et al. 2009) and pre-digested at room temperature for at least 30 minutes before initiating microwave-assisted acid digestion (Kuss 1992). All samples, CRMs, and blanks were digested completely (solubilizing all solid material) using the MARS 6 Digestion Microwave System (CEM Corporation, Matthews, North Carolina, U. S. A.) by stepwise ramping up the temperature from room temperature to 200° C for 15 min and holding at 200° C for 10 minutes to ensure complete digestion. After cooling, approximately 1 mL of diluted samples were transferred to auto-sampling vials in preparation for instrumental analysis.

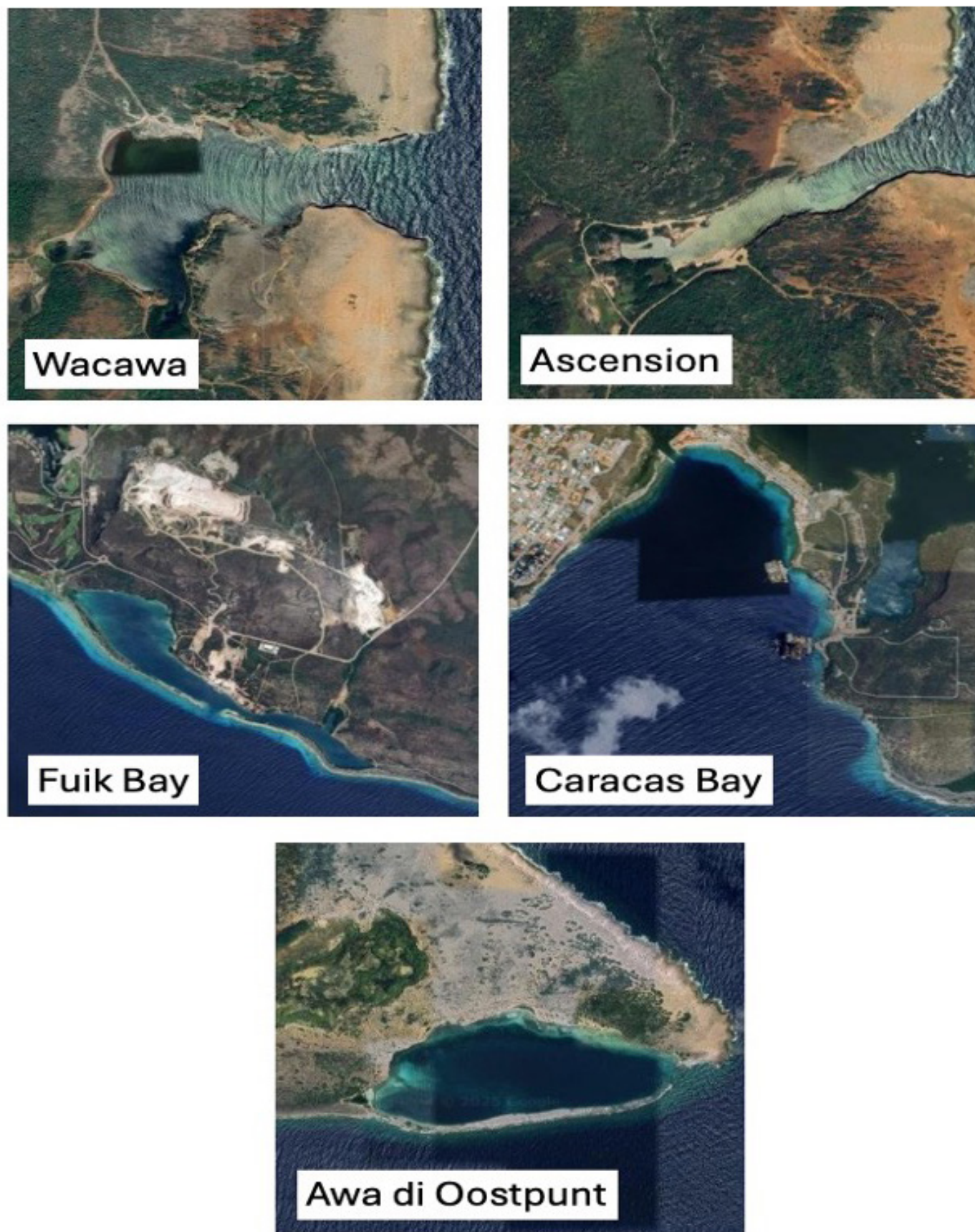


FIGURE 2. Aerial images of each collection site showing differences in adjacent land usage (Google Earth).

TABLE 1. Summary of the number of samples, common names, and scientific names for each of the five locations of sample collection.

Collection Site	Number of Samples	Common Name	Scientific Name
Ascension	9	green feather algae (seaweed)	<i>Caulerpa sertularioides</i>
Awa di Oostpunt	2	manatee seagrass	<i>Syringodium filiforme</i>
Awa di Oostpunt	2	halophila seagrass	<i>Halophila stipulacea</i>
Awa di Oostpunt	2	turtle seagrass	<i>Thalassia testudinum</i>
Awa di Oostpunt	2	green feather algae (seaweed)	<i>Caulerpa sertularioides</i>
Caracas Bay	6	halophila seagrass	<i>Halophila stipulacea</i>
Fuik Bay	6	halophila seagrass	<i>Halophila stipulacea</i>
Wacawa	3	green feather algae (seaweed)	<i>Caulerpa sertularioides</i>
Wacawa	2	turtle seagrass	<i>Thalassia testudinum</i>
Wacawa	2	unknown	unknown
Wacawa	2	green feather algae (seaweed)	<i>Caulerpa sertularioides</i>

Instrumental Analysis

Concentrations of copper (an essential metal) and lead (a non-essential metal) were quantified using a Graphite Furnace Atomic Absorption Spectrometer (GFAAS) (Agilent, Model 120, U. S. A.) with an autosampler. Stock standards of each metal were prepared from reference standard solutions certified at 1000 ppm $\pm$ 1% (Fisher Chemical). For each metal, a five-point external calibration was used via the GFAAS creating the dilution series from stocks solutions with dilution with ultra-trace elemental analysis grade water (Fisher Chemical, U. S. A.). Twenty microliter aliquots of each sample were injected into a room temperature graphite tube installed within the GFAAS. Default temperature ramps for the two metals were used with the maximum temperature for atomic absorption at around 2600° C Celsius. The wavelengths for monitoring absorption for copper and lead were 324.8 and 283.3 nm, respectively, using a slit width of 0.5 nm. No background correction or matrix modification were employed. Peak height was used to quantify the concentration of metals in each sample. Concentrations were reported as parts per billion (ppb), that is ug/g, expressed on a dry weight of sample (dw) basis.

Quality Control and Quality Assurance

For quality control and method assessment, a plankton certified reference material (CRM) (BCR®-414; European Commission, Joint Research Center) was used in triplicate analysis. At a frequency of 10%, blank samples consisting of only 10.0 mL of HNO₃ and triplicate analyses of four randomly chosen sea grass samples were processed. Blanks, triplicate analyses, and CRMs were extracted and analyzed using identical methods for samples, as outlined above. Triplicate analyses of samples were only used to assess precision, meaning those replicated values were not averaged with the original samples.

Statistical Analysis

Kruskal-Wallis tests were performed to identify if significant differences existed in the concentrations of copper and lead in locations and species of seagrass. Dunn's test using the Benjamini-Hochberg method was then used to identify differences between pairs of two groups (Dunn 1961). A Kruskal-Wallis test is suitable for comparing differences between three or more groups when sample sizes are small and the assumptions for an analysis of variance test are not realistic (Kruskal 1957). Dunn's test provides a non-parametric method to compute pairwise comparisons of groups following a Kruskal-Wallis test (Dinno 2015). All statistical analyses were performed in R.

TABLE 2. Summary of mean concentrations (ppb, dw) of copper and lead in seagrasses and seaweeds from five sites.

Site	Copper (mean $\pm$ standard deviation of the mean)	Lead (mean $\pm$ standard deviation of the mean)
Ascension (AS)	3.68 $\pm$ 1.24	1.70 $\pm$ 0.74
Awa di Oostpunt (AO)	2.38 $\pm$ 1.46	0.60 $\pm$ 0.18
Caracas Bay (CB)	5.86 $\pm$ 2.03	2.83 $\pm$ 0.61
Fuik Bay (FB)	4.20 $\pm$ 1.60	1.35 $\pm$ 0.35
Wacawa (WA)	4.59 $\pm$ 2.73	0.94 $\pm$ 0.46

RESULTS

Qualifying Data

Calibration curves for copper and lead were highly linear ($r^2 > 0.98$) in the linear dynamic range used to quantify the concentrations in all samples, CRMs and blanks. For those samples falling out of the linear dynamic range, dilution using ultra-trace elemental analysis grade water (Fisher Chemical, U. S. A.) was performed. Using the blanks, the instrumental detection limit for copper and lead, determined using the mean plus 3 times the standard deviation of the mean, was 0.11 and 0.29 ppb (dw), respectively. All samples had concentrations above the instrumental detection limits for the two metals.

The precision of the analytical and instrumental protocols used was assessed using triplicate analyses of random samples. From the four randomly chosen samples used for triplicate analysis, the percent coefficient of variation (defined as the standard deviation of mean divided by the mean expressed as a percentage) across all replicates was 20% for copper and 25% for lead. This was indicative of a very good level of precision considering the relatively low concentrations (single digit values for ppb) that the GFAAS is capable of accurately measuring.

The accuracy of the analytical and instrumental protocols was assessed using the results of the CRM triplicate analysis. Recoveries for each triplicate analysis were determined by the commonly acceptable practice of dividing the concentration of the metal determined by the certified value and expressing it as a percentage (Afridi et al. 2009). The mean recoveries for copper and lead were 83% and 123% for copper and lead, respectively. For copper and lead, the acceptable range for CRM recoveries is typically 15% to 20% $\pm$ the certified values (Sutherland and Tack 2002; USDA

2021), though this does depend on metal analyte, extraction method, instrument employed, and on the matrix of the CRM used (Hassan et al. 2007; Lee et al. 2022). While copper fell in this range by 2%, lead was slightly higher by 3%. However, on average, between the two metals, our recoveries were 20% $\pm$ the certified values, within the range of acceptability and denoting a sufficient level of accuracy in the analytical and instrumental protocols. Thus, the concentrations reported in this study reflect close to true values existing in the samples. Reported concentrations were not adjusted for CRM recovery, meaning the concentrations reported may be considered, on average, to be around 20% less than the true values for copper and 20% larger for lead.

Summary of Concentrations

Copper and lead were found in all samples, above detection limits. Concentrations of copper and lead, reported in ppb (dw), were averaged to determine the mean, and a standard deviation of the mean was calculated (Table 2). Across the five sites, mean concentrations of copper were correlated to mean concentrations of lead (linear regression, $r^2 = 0.65$). That is, sites that had higher concentrations of copper generally had higher concentrations of lead. At each station, the mean copper concentrations were higher compared to lead concentrations.

Copper concentrations, reported as ppb (dw), varied between the five sites (Table 2; Fig. 3). Awa di Oostpunt had the lowest mean copper concentration, while Caracas Bay had the highest mean copper concentration. Ascension Bay, Fuik Bay, and Wacawa had similar copper concentrations that were intermediate between Caracas Bay and Awa di Oostpunt.

As observed with copper, lead concentrations, reported in ppb (dry wgt), varied between the five sites (Table 2; Fig. 4). As with copper, Awa di Oostpunt had

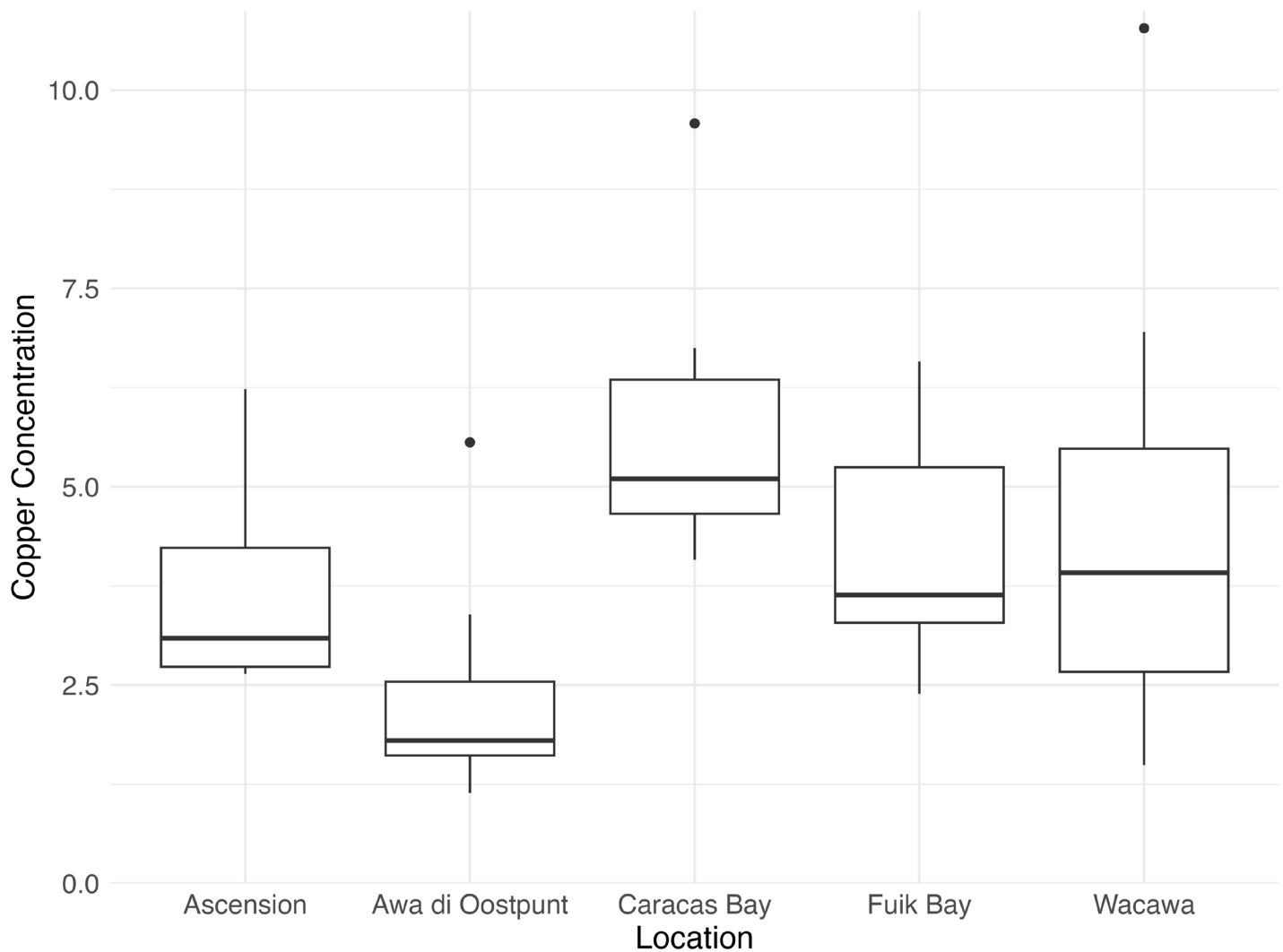


FIGURE 3. Box Plot for the concentrations (ppb, dw) of copper in seagrasses from the five locations. Boxes represent the interquartile range (25–75% of observations), with the median shown as the horizontal line. Whiskers extend to the most extreme values within $1.5 \times \text{IQR}$ below the first quartile and above the third quartile. Values beyond the whiskers are plotted as outliers.

the lowest mean lead concentration while Caracas Bay had the highest. Ascension Bay, Fuik Bay, and Wacawa had similar mean lead concentrations that were intermediate between Caracas Bay and Awa di Oostpunt.

Kruskal-Wallis tests were employed for both metals, each with locations group and species group. The null hypothesis states there are no differences between groups while the alternative hypothesis states that at least two groups differ. The alternative hypothesis was rejected in every case except with copper metal and species group. There were significant differences in copper concentrations between at least two locations (p value = 0.015), in lead concentrations between at least two locations (p value = 0.00003), and in lead concentrations between at least two species (0.008). To iden-

tify the significant differences between pairs of two groups, Dunn's tests using the Benjamini-Hochberg method were used. For lead concentrations, there were only significant differences between *Caulerpa sertularioides* and *Thalassia testudinum* (at 0.05 confidence level) and between *Halophia stipulacea* and *Thalassia testudinum* (at 0.01 confidence level). For lead concentrations, there were significant differences between all sites (at varying confidence levels) except between Fuik Bay and Ascension (Table 3). For copper concentrations, there were significant differences (at varying confidence levels) between Awa di Oostpunt and each of the four other sites only, and between Ascension and Caracas Bay (Table 4).

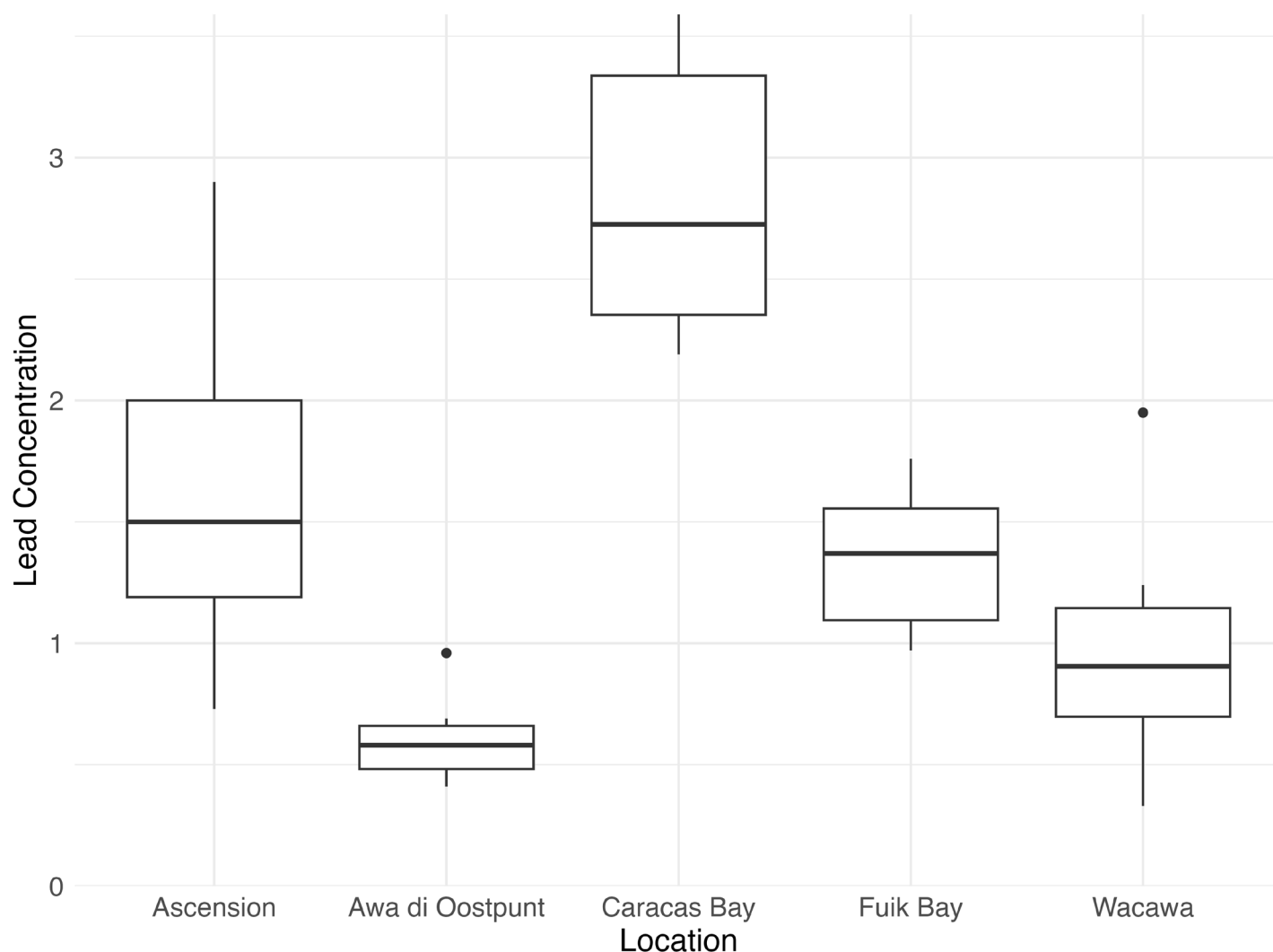


FIGURE 4. Box Plot for the concentrations (ppb, dw) of lead in seagrasses from the five locations. Boxes represent the interquartile range (25–75% of observations), with the median shown as the horizontal line. Whiskers extend to the most extreme values within $1.5 \times \text{IQR}$ below the first quartile and above the third quartile. Values beyond the whiskers are plotted as outliers.

DISCUSSION

Spatial Differences in Copper and Lead Concentrations

This study found no significant differences in copper concentrations based on different species. For lead concentrations, there were only significant differences between *Caulerpa sertularioides* and *Thalassia testudinum* and between *Halophia stipulacea* and *Thalassia testudinum*. This suggests that the differences observed in metal concentrations were not largely driven by species for lead and not at all for copper. However, for both metals, concentrations significantly differed based on location, indicating that variable copper and lead inputs between sites were likely driving most of the con-

centration differences observed in samples. Also, copper and lead levels were well correlated to each other at most sites (that is, a site that had higher lead concentrations often had higher copper concentrations), suggesting a common source and/or delivery mechanism for both metals.

Considering all five sites, Caracas Bay harbored the highest copper and lead seagrass/seaweed concentrations, while Awa di Oostpunt had the lowest. The three other sites had concentrations lying between these two sites for both metals. As mentioned previously, a suite of processes may deliver trace metals to coastal environments. Regional environmental conditions, such as ocean currents, sediment composition, and anthropogenic activity, may uniquely shape metal contamination.

TABLE 3. Dunn's test for pairwise comparison, using the Benjamini-Hochberg method for lead concentrations and location. Cell values show column mean-row mean.

	Awa di Oostpunt	Ascension	Caracas Bay	Fuik Bay
Ascension	-3.4***			
Caracas Bay	-4.6****	-1.6*		
Fuik Bay	-2.6**	0.5	1.9*	
Wacawa	-1.4*	2.2**	3.5**	1.5*

*|**|***|**** show significance at 0.10, 0.05, 0.01, and 0.001 levels respectively.

TABLE 4. Dunn's test for pairwise comparison, using the Benjamini-Hochberg method for copper concentrations and location. Cell values show column mean-row mean.

	Awa di Oostpunt	Ascension	Caracas Bay	Fuik Bay
Ascension	-1.8*			
Caracas Bay	-3.4***	-1.8*		
Fuik Bay	-2.1*	-0.5	1.2	
Wacawa	-2.4**	-0.6	1.4	0.03

*|**|***|**** show significance at 0.10, 0.05, 0.01, and 0.001 levels respectively.

tion patterns between locations (Fuge and James 1973). However, most studies using seagrass as bioindicators have found that adjacent urbanized/industrialized land use and waste discharges, or riverine inputs, are likely the main sources of metals to the aquatic environment. Farias et al. (2018) evaluated the concentration of As, Cu, Pb, and Zn in macrophytes along the Derent estuary (Tasmania), finding the highest levels in the historically contaminated area of the estuary, suggesting that legacy contamination with the sediments could also constitute a route of uptake for seagrasses. With no known legacy contamination in the sediments of coastal bays in Curaçao, and no significant riverine inputs on the island, potential land-based differences could explain spatial differences in copper and lead in seagrasses.

The eastern part of Curaçao is relatively undeveloped compared to other parts of the island (Debrot, et al. 2017). Awa di Oostpunt lies at the eastern tip of the island and is considered very remote with no visible or known potential land-based inputs of trace metals. Not surprisingly, the lowest concentrations of both metals in seagrasses were observed there. According to Bertoneclj et al. (2025), who modelled particle distribution around Curaçao, most land-based particles, and likely particle-associated contaminants, reach this embayment through currents from Klein Curaçao, an unpopulated island. Those particles are not likely to be associated with elevated levels of trace metals due to their remoteness

of origin and lack of anthropogenic activities. Moreover, they suggest that the area encompassing Awa di Oostpunt has more water renewal compared to other parts of Curaçao, meaning the embayment waters are likely flushed more regularly, potentially decreasing exposure to trace metals (Bertoneclj, unpublished data). However, it is possible, and should be considered in the future, that large riverine inputs from Central American and northern South American countries (Guzmán and Jiménez 1992) may deliver trace metal bound sediments that could potentially be dispersed to southern Caribbean islands such as Curaçao.

Caracas Bay represents the most urbanized/industrialized site in this study, and because of this, its highest copper and lead levels were not surprising. It is unlikely that particle-bound metals transferred by currents to this area are influencing seagrass metal inventories. Through particle modelling, Bertoneclj et al. (2025) have shown that Caracas Bay and Fuik Bay receive most current-transferred particles from the easternmost point of the island, where Awa di Oostpunt and other less developed areas are located. These areas have little anthropogenic footprint, and likely particles are not laden with elevated trace metals. Caracas Bay is likely receiving trace metal inputs from both wastewater and industrial discharges. Caracas Bay, along with its neighbor Fuik Bay, were found to have concerning levels of sewage and trash indexes

according to the Waitt Institute's (2017) publication on "The State of Curaçao's Coral Reef." Moreover, in reference to coral reef health and integrity, as highlighted by Snijders (2023), one issue of concern was Curaçao's wastewater management protocols that result in high amounts of sources of pollution via sewage system effluent discharges into coastal systems. While there are four wastewater treatment plants on the island, Snijders (2023) reported that they have widely varying capacities and efficiencies. He reported that only 33% of Curaçao's population is connected to wastewater treatment plants, which are outdated and insufficient in their effective treatments. Industrial wastes entering coastal ecosystems may also explain the differences in trace metal concentrations between sites. Snijders (2023) reported that the Curaçaoan government states that at least 90% of all industrial wastewater is discharged either directly into the ocean or into the sewer system. Discharge sites into coastal waters are highly concentrated around the southwest portion of the island adjacent to the more heavily populated center of Williamstad (Snijders 2023). Caracas Bay seagrasses and seaweeds may be reflecting the copper and lead levels within these wastewater and industrial discharges. However, additional studies monitoring seagrass/seaweed metal concentrations and other potential contaminants adjacent to known discharge sites are recommended to investigate these as significant delivery mechanisms.

The three sites having moderate levels of copper and lead (Fuik Bay, Wacawa, Ascension) have variable adjacent land-use features that potentially deliver trace metals to the seagrass communities. Fuik Bay's adjacent land use is dominated by a limestone mining operation on its top and hillside. Once quarried, limestone reserves are crushed, often creating smaller particles and dust that can be distributed to local soils in and around the operation. Past studies have shown limestone mining can enrich soils in trace metals. For example, Mulwa et al. (2012) found various heavy metals, including copper and lead, in both limestone and soil samples in limestone deposits from Kenya. Fuik Bay lies in the entire watershed of the top and hillside mining operation. Although the arid climate of Curaçao likely means a reduced run-off potential of trace metals, it is possible that dust and soils containing copper and lead may enter the embayment through runoff during rainy season

events and through aeolian transport during dry and windy conditions.

Wacawa currently is used a military training and shooting range. Past research has identified higher levels of lead at or near shooting ranges (Cao et al. 2003; Shaw et al. 2023) and lead mobility in soils has been documented (Murray et al. 1997). However, lead concentrations in Wacawa-adjacent seagrasses and seaweeds in this study were relatively low compared to other sites. In fact, lead concentrations were the second lowest observed between the 5 sites, and rivaled the low levels observed at the most remote site of Awa di Oostpunt. From our limited dataset, it does not appear that the military training and shooting range has a discernable impact on copper or lead accumulation in seagrasses. However, a limited number of samples were analyzed, during only one sampling event. Further studies should expand the number of locations, species, and individual samples to more fully assess the role of adjacent land uses at this site. Moreover, evaluation of the speciation of lead entering and within the water of the bay and within the sediments would assist in determining the concentrations of the bioavailable form to seagrasses.

Ascension had similar copper levels to Fuik Bay and Wacawa but had the second highest lead concentrations. As noted by Govers et al. (2014), the upper bay is used for trash and waste disposal. If this waste includes electronic components, that may be a source for copper and lead to the aquatic ecosystem. Again, further larger scale studies should be conducted.

In summary, though speculation on significant and potential sources of trace metals has been proposed to explain spatial differences based on the statistically significant differences in metal concentrations between sites, differences between sites may be characterized as minimal. Moreover, concentrations ranged only in the single digit ppb range, with no site having higher concentrations beyond this range. That is, no 'hot spot' area of concern regarding potential contamination was identified.

Comparison to Other Studies

For decades, trace metal concentrations in seagrasses and seaweeds have been widely reported in the literature. However, caution should be taken when comparing previously published concentrations for several reasons. Firstly, researchers concluded that concentra-

TABLE 5. Comparison of copper and lead concentration in various seagrass species worldwide.

Seagrass Species	Study	Location	Copper Concentrations (ppb dw)	Lead Concentrations (ppb dw)
Various species (above ground tissues)	This Study	Curaçao	2.38 to 5.86 (range of means from 5 sites)	0.60 to 2.83 (range of means from 5 sites)
<i>Thalassia testudinum</i>	Govers et al. 2014	Curaçao and Bonaire	1.36 to 31.0	0.97 to 5.93
<i>Zostera japonica</i> (above ground tissues)	Lin et al. 2016	Yellow River Estuary, China	18.0 (mean from 3 sites)	37.3 (mean from 3 sites)
<i>Posidonia oceanica</i>	Bonanno and Di Marino 2017	Sicily, Italy	11.6 ± 1.9	3.22 ± 0.42
<i>Enhalus acoroides</i> (leaves)	Jeong et al. 2021	Palau	0.9 to 6.7 (range)	0.1 to 2.3 (range)
Species consumed by <i>Chelonia mydas</i>	Wilkinson et al. 2022	Great Barrier Reef, Australia	2.47 to 5.64 (range)	na
<i>Zostera japonica</i> (above ground tissue)	Wang et al. 2025	Yellow Sea, China	6.13 to 7.83 (range of means from 4 sites)	0.52 to 3.01 (range of means from 4 sites)

tions may vary based on the species (Farias et al. 2018; Bibak et al. 2020; El-Seifat et al. 2023), regardless of the ambient metal concentrations in the water and sediment. For example, El-Seifat et al. (2023) compared seven different species of marine macroalgae from six stations in coastal Libya and found *Laurencia* species exhibited the highest abilities to concentrate metals. Secondly, concentrations in a single species may vary depending upon whether the blade, stem, roots, or whole plant were analyzed (Schalacher-Hoenlinger and Schalacher 1998). For example, using *Posidonia oceanica* seagrass as a biomonitoring tool in coastal Tunisia, El Zrelli et al. (2017) observed enhanced copper accumulation in leaves as compared to roots, while lead was concentrated to a higher degree in roots. Others have found greater trace metal accumulation in above-ground tissue (leaves and stems) than underground tissue (roots) (Wang et al. 2025). Thirdly, reported concentrations reflect contaminant inventories at the time of sampling. Schalacher-Hoenlinger and Schalacher (1998) observed significant variability in concentrations based on time of sampling (seasonal variability). Lastly, while it is standard to report concentrations on a dry weight basis, some studies fail to state whether con-

centrations were normalized to dry weight of seagrass or seaweed and reported as such.

Despite these caveats, comparing this study's dataset with those collected from other regions of the world is potentially useful in identifying any significantly elevated concentrations. To this end, six recent (past 11 years) studies evaluating copper and lead concentrations in various seagrass species worldwide were summarized (Table 5). Except for one study evaluating Yellow River Estuary seagrasses (Lin et al. 2016), copper concentrations (either means or ranges) were generally higher than lead concentrations, mirroring this study, and may be indicative of copper's role as a micronutrient for growth at low environmental concentrations. Again, apart from the Lin et al. (2016) study, means or ranges for the two metals were quite similar across all studies. This is likely because most, if not all, of these studies evaluated seagrasses from locations that could be considered not significantly impacted by major contaminant inputs. Although this limited comparison reveals that the highest concentrations were from the Yellow River Estuary, Lin et al. (2016) indicated that heavy metals in seawater from their sampling locations were relatively low, indicating good water quality.

Moreover, their ecological risk assessments found no significant pollution risk at the levels of copper, lead, and other metals reported. Our study's range of means from five sites was not very different from those found worldwide, even when considering that different species were evaluated. This suggests that copper and lead concentrations presented here are likely not at levels of concern for ecological risk, though laboratory-based toxicological studies would be recommended to support this (Prange and Dennison 2000).

Potential Trophic Transfer of Metals to Sea Turtles

Metals sequestered in the leaves, shoots, and roots of seagrasses and seaweed may be transferred to aquatic organisms that feed on these living or decaying species. This trophic transfer may expose higher organisms to trace metals depending on the concentrations within the ingested tissue. However, unlike arsenic and mercury, most trace metals such as copper and lead have not been shown to biomagnify in aquatic food webs (Schneider et al. 2018). That is, there is no evidence to support increased concentrations of metals with increasing trophic levels.

Seagrasses are fed upon by a suite of herbaceous and omnivorous marine organisms such as fish (Prado and Heck 2011), sea urchins (Klump et al. 1993), and green sea turtles (Mortimer 1981; Estaban et al. 2020). If dietary items such as seagrasses and seaweeds have elevated levels of trace metals, or other contaminants, bioaccumulation may occur (Caurant et al. 1999) and deleterious effects in the consumers may incur. One such effect that is currently being evaluated is the role of contaminants in potentially increasing green sea turtle susceptibility to fibropapillomatosis (FP).

FP is a disease that causes benign tumors in sea turtles (Jones et al. 2016). Though not lethal, depending on where the tumors grow, they decrease sight, mobility, and foraging (Aguirre and Lutz 2004) reducing the health of the turtle. Researchers have suggested environmental pollutants may play a role in contributing to greater incidences of FP by suppressing sea turtles' immune systems, allowing greater susceptibility to the herpesvirus causing the disease (Herbst 1994). However, the list of potential pollutants is enormous, making evaluations and links incredibly daunting for scientists. Keller et al. (2014) studied 164 persistent organic pollutants in 52 Hawaiian green sea turtles and concluded their data did not suggest a major cofactor in causing the

onset of FP. However, other pollutants, such as heavy metals, may (Godley et al. 1999; Carneiro et al. 2016).

Recently, Tripepi et al. (2025) used telemetry in Caracas Bay to track both healthy and FP-affected green sea turtles. They found high site-fidelity for seagrass foraging within that embayment. As noted in this study, the highest concentrations of copper and lead were found at Caracas Bay. However, concentrations were not vastly different from other sites, including those considered to be remote and pristine, nor were they sufficient to indicate a 'hot spot' for these two metals. At these levels, it is likely that copper and lead exposure to green sea turtles via trophic transfer of ingested seagrasses and seaweeds is not a cofactor in FP susceptibility. However, more research is warranted to fully assess the potential role of these trace metals, and other contaminants, as vectors for increasing susceptibility to FP.

Directions for Future Studies

With five sites evaluated and a total of 38 seagrass/seaweed samples analyzed, this study may be considered an important first step in more fully evaluating potential trace metal contamination in Curaçaoan coastal ecosystems. The inclusion of additional sites around the island, especially those adjacent to known wastewater discharges (and illegal co-discharges of industrial waste), would allow for a better understanding of the concentrations in seagrasses along the entire coast of the country, and increase the spatial resolution in mapping contaminant source, transport, and fate to more accurately pinpoint potential land-based discharges of wastewater or industrial releases. In addition, augmenting seagrass and seaweed concentrations with sediment and water concentrations would be helpful in assessing the route of trace metal exposure to these coastal species.

Curaçao's combination of low annual rainfall, location outside the hurricane belt, and relatively flat topography, likely does not deliver large and continuous influxes of runoff containing land-based pollutants. However, Curaçao receives its peak rain intensities in December, when short downpours may deliver enough rain and runoff to cause challenges to efficient surface runoff drainage management (Snijders, 2023). Future studies should focus on the importance of these seasonal and episodic events in delivering trace metal bound particles to coastal ecosystems.

CONCLUSIONS

This study, although limited in spatial and temporal scope, provides essential contaminant data for identifying major inputs and processes as well as for environmental health assessment. For Curaçao, although significant differences in copper and lead concentrations in seagrasses between sites may arise from wastewater and industrial discharges, elevated levels of concern were not observed. However, more extensive data, at finer regional scales and more frequent time intervals, is needed to more fully assess the health of coastal ecosystems in Curaçao and the WCR. As approximately one of every four jobs within the WCR depends on healthy marine ecosystems (De Paulo 2016), the ecological health and economic importance of these ecosystems represent vital factors in justifying the need for more monitoring studies in Curaçao's coasts.

Acknowledgements—This work was funded in part by Thomas Jefferson University through a foundation grant provided by the Eileen Martinson '86 Fund.

LITERATURE CITED

- Afridi, H. I., T. G. Kazi, N. G. Kazi, M. K. Jamali, R. A. Sarfaraz, M. B. Arain, G. A. Kandhro, A. Q. Shah, J. A. Baig, N. Jalbani, and R. Ansari. 2009. Determination of copper and iron in biological samples of viral hepatitis (A-E) female patients. *Biological trace element research* 129: 78–87.
- Aguirre, A. A. and P. L. Lutz. 2004. Marine Turtles as Sentinels of Ecosystem Health: Is Fibropapillomatosis an Indicator? *EcoHealth* 1: 275–283.
- Al Naggar, Y., M. S. Khalil, and M. A. Ghorab. 2018. Environmental Pollution by Heavy Metals in the Aquatic Ecosystems of Egypt. *Open Access Journal of Toxicology* 3: 1–9.
- Baptista Nobre, T., L. S. Wright, D. Kneer, D. Priosambodo, and S. C. A Ferse. 2024. Trace metal pollution gradients in a tropical seagrass ecosystem. *Marine Environmental Research* 200: 1–13.
- Bertoncelj, V., F. Mienis, P. Stocchi, and E. van Sebile. 2025. Flow patterns, hotspots, and connectivity of land-derived substances at the sea surface of Curaçao in the southern Caribbean. *Ocean Science* 21: 935–964.
- Bibak, M., M. Sattari, S. Tahmasebi, A. Agharokh, and J. I. Namin. 2020. Marine macro-algae as a bio-indicator of heavy metal pollution in the marine environments, Persian Gulf. *Indian Journal of Geo Marine Sciences* 49: 357–363.
- Bonanno, G. and V. Di Martino. 2017. Trace element compartmentation in the seagrass *Posidonia oceanica* and biomonitoring applications. *Marine Pollution Bulletin* 116: 196–203.
- Cao, X., L. Q. Ma, M. Chen, D. W. Hardison Jr., and W. G. Harris. 2003. Lead transformation and distribution in the soils of shooting ranges in Florida, USA. *Science of The Total Environment* 307: 179–189.
- Carneiro, C., R. D. Klein, I. F. Barcarolli, and A. Bianchini. 2016. Metal contamination as a possible etiology of fibropapillomatosis in juvenile female green sea turtles *Chelonia mydas* from the southern Atlantic Ocean. *Aquatic Toxicology* 170: 42–51.
- Caurant, F., P. Bustamante, M. Bordes, and P. Miramand. 1999. Bioaccumulation of cadmium, copper and zinc in some tissues of three species of marine turtles stranded along the French Atlantic coasts. *Marine Pollution Bulletin* 38: 1085–1091.
- Debrot, D., J. Langley, M. Vermeij, D. Slijkerman, R. Jongbloed, I. van Beek, J. de Freitas, L. Pors, R. Jak, C. Rockmann, and J. Cremer. 2017. Description of the Outstanding Universal Value (OUV) of the Proposed Marine Nomination Properties of the Bonaire and Curaçao Marine Parks (BCMP). Available at: <https://edepot.wur.nl/430905>.
- De Paulo, H. 2016. Definitely not a waste of time. Available at: <https://blocks.mvmm.nl/images/pdf/8404.pdf>.
- Diez, S. M., P. G. Patil, J. Morton, D. J. Rodriguez, A. Vanzella, D. V. Robin, T. Maes, and C. Corbin. 2019. *Marine Pollution in the Caribbean: Not a Minute to Waste*. World Bank Group.
- Dinno, A. 2015. Nonparametric pairwise multiple comparisons in independent groups using Dunn's test. *Stata Journal* 15: 292–300.
- Duce, R. A., P. S. Liss, J. T. Merrill, E. L. Atlas, P. Buat-Menard, B. B. Hicks, J. M. Miller, J. M. Prospero, R. Arimoto, T. M. Church, W. Ellis, J. N. Galloway, L. Hansen, T. D. Jickells, A. H. Knap, K. H. Reinhardt, B. Schneider, A. Soudine, J. J. Tokos, S. Tsunogai, R. Wollast, and M. Zhou. 1991. The atmospheric input of trace species to the world ocean. *Global Biogeochemical Cycles* 5: 193–259.
- Dunn, O. J. 1961. Multiple comparisons among means. *Journal of the American Statistical Association* 56:

- 52–64.
- El-Seifat, R. M., S. M. Bufarwa, M. F. Gaballa, M. A. A. Soutiyah, S. A. Abdulhafith, A. Saleh, V. P. Dhulap, K. D. Ahire, and I. H. Attitalla. 2023. The Role of Marine Algae as a Bioindicator in Assessing Environmental Pollution. *Journal of Survey in Fisheries Sciences* 10: 1837–1869.
- El Zrelli, R., P. Courjault-Radé, L. Rabaoui, N. Daghouj, L. Mansour, R. Balti, S. Castet, F. Attia, S. Michel, and N. Bejaoui. 2017. Biomonitoring of coastal pollution in the Gulf of Gabes (SE, Tunisia): use of *Posidonia oceanica* seagrass as a bioindicator and its mat as an archive of coastal metallic contamination. *Environmental Science and Pollution Research* 24: 22214–22225.
- Esteban, N., J. A. Mortimer, H. J. Stokes, J.-O. Laloë, R. K. F. Unsworth, and G. C. Hays. 2020. A global review of green turtle diet: sea surface temperature as a potential driver of omnivory levels. *Marine Biology* 167: 183 1–17.
- Farias, D. R., C. L. Hurd, R. S. Eriksen, and C. K. Macleod. 2018. Macrophytes as bioindicators of heavy metal pollution in estuarine and coastal environments. *Marine Pollution Bulletin* 128: 175–184.
- Fernandez, A., A. Singh, and R. Jaffé. 2007. A literature review on trace metals and organic compounds of anthropogenic origin in the Wider Caribbean Region. *Marine Pollution Bulletin* 54: 1681–1691.
- Fuge, R. and K. H. James. 1973. Trace metal concentrations in brown seaweeds, Cardigan Bay, Wales. *Marine Chemistry* 1: 281–293.
- Godley, B. J., D. R. Thompson, and R. W. Furness. 1999. Do heavy metal concentrations pose a threat to marine turtles from the Mediterranean Sea? *Marine Pollution Bulletin* 38: 497–502.
- Gonzalez, M. H., G. B. Souza, R. V. Oliveira, L. A. Forato, J. A. Nóbrega, and A. R. A. Nogueira. 2009. Microwave-assisted digestion procedures for biological samples with diluted nitric acid: Identification of reaction products. *Talanta* 79: 396–401.
- Govers, L. L., L. P. M. Lamers, T. J. Bouma, J. H. F. de Brouwer, and M. M. van Katwijk. 2014. Eutrophication threatens Caribbean seagrasses—An example from Curaçao and Bonaire. *Marine Pollution Bulletin* 89: 481–486.
- Guyvenchy, F., L. Vornicu, L. Șmuleac, and R. Pașcălu. 2023. Pollution and Biodiversity in the Caribbean. *Research Journal of Agricultural Science* 55: 72–77.
- Guzmán, H. M. and C. E. Jiménez. 1992. Contamination of coral reefs by heavy metals along the Caribbean coast of Central America (Costa Rica and Panama). *Marine Pollution Bulletin* 24: 554–561.
- Hardy, D. R., C. A. Niño Torres, A. Pereira Corona, and M. d. P. Blanco Parra. 2025. Potentially toxic elements in the Caribbean coastal region: A review. *Revista Internacional de Contaminación Ambiental* 41: 311–339.
- Hassan, N. M., P. E. Rasmussen, E. Dabek-Zlotorzynska, V. Celo, and H. Chen. 2007. Analysis of Environmental Samples Using Microwave-Assisted Acid Digestion and Inductively Coupled Plasma Mass Spectrometry: Maximizing Total Element Recoveries. *Water, Air, and Soil Pollution* 178: 323–334.
- Herbst, L. H. 1994. Fibropapillomatosis of marine turtles. *Annual Review of Fish Diseases* 4: 389–425.
- Hornbach, M. J., P. J. Mann, F. Taylor, S. W. Bowen, and J. J. Pickle. 2010. Estimating the Age of Near-Shore Carbonate Slides Using Coral Reefs and Erosional Markers: A Case Study from Curacao, Netherlands Antilles. *The Sedimentary Record* 8: 4–10.
- Janssen, J. A. M., E. A. T. Houtepen, A. S. J. van Proosdij, T. van Steveninck, R. Haveman, and I. de Ronde. 2023. Vegetatiekartering en inventarisatie witstaarthert op militair oefen- en schietterrein Wacawa (Curaçao). *Wageningen Environmental Research* 3255: 1–51.
- Jeong, H., J. Y. Choi, D.-H. Choi, J. -H. Noh, and K. Ra. 2021. Heavy metal pollution assessment in coastal sediments and bioaccumulation on seagrass (*Enhalus acoroides*) of Palau. *Marine Pollution Bulletin* 163: 111912.
- Jones, K., E. Ariel, G. Burgess, and M. Read. 2016. A review of fibropapillomatosis in Green turtles (*Chelonia mydas*). *The Veterinary Journal* 212: 48–57.
- Joshi, U. M. and R. Balasubramanian. 2010. Characteristics and environmental mobility of trace elements in urban runoff. *Chemosphere* 80: 310–318.
- Kasongo, J., L. Y. Alleman, J.-M. Kanda, A. Kaniki, and V. Riffault. 2024. Metal-bearing airborne particles from mining activities: A review on their characteristics, impacts and research perspectives. *Science of The Total Environment* 951: 175426.

- Keller, J. M., G. H. Balazs, F. Nilsen, M. Rice, T. M. Work, and B. A. Jensen. 2014. Investigating the Potential Role of Persistent Organic Pollutants in Hawaiian Green Sea Turtle Fibropapillomatosis. *Environmental Science & Technology* 48: 7807–7816.
- Klumpp, D. W., J. T. Salita-Espinosa, and M. D. Fortes. 1993. Feeding ecology and trophic role of sea urchins in a tropical seagrass community. *Aquatic Botany* 45: 205–229.
- Kolarova, N. and P. Napiórkowski. 2021. Trace elements in aquatic environment. Origin, distribution, assessment and toxicity effect for the aquatic biota. *Ecohydrology & Hydrobiology* 21: 655–668.
- Kruskal, W. H. 1957. Use of ranks in one criterion variance analysis. *Journal of the American Statistical Association* 52: 8–21.
- Kuss, H.-M. 1992. Applications of microwave digestion technique for elemental analyses. *Fresenius' Journal of Analytical Chemistry* 343: 788–793.
- Lee, K. J., E. H. Kang, M. Yoon, M. R. Jo, H. S. Yu, and K. T. 2022. Concentration of heavy metals in shellfishes and health risk assessment from Korean coastal areas. *Fisheries and Aquatic Sciences* 25: 626–636.
- Lin, H., T. Sun, S. Xue, and X. Jiang. 2016. Heavy metal spatial variation, bioaccumulation, and risk assessment of *Zostera japonica* habitat in the Yellow River Estuary, China. *Science of The Total Environment* 541: 435–443.
- Martínez-Crego, B., A. Vergés, T. Alcoverro, and J. Romero. 2008. Selection of multiple seagrass indicators for environmental biomonitoring. *Marine Ecology Progress Series* 361: 93–109.
- Marrugo-Negrete, J., J. Pinedo-Hernández, S. Marrugo-Madrid, E. Navarro-Frómata, and S. Díez. 2021. Sea Cucumber as Bioindicator of Trace Metal Pollution in Coastal Sediments. *Biological Trace Element Research* 199: 2022–2030.
- Mulwa, B. M., D. M. Maina, and J. P. Patel. 2012. Multielemental analysis of limestone and soil samples of Kitui South (Kenya) limestone deposits. *International Journal of Fundamental Physical Sciences* 2: 48–51.
- Mishra, A. K., R. Sahoo, S. S. Samantaray, and D. Apte. 2022. Seagrass Ecosystems of India as Bioindicators of Trace Elements. Pp. 45–65 in *Coastal Ecosystems*. S. Madhav, S. Nazneen, and P. Singh (eds.). Coastal Research Library, vol 38. Springer.
- Mortimer, J. A. 1981. The Feeding Ecology of the West Caribbean Green Turtle (*Chelonia Mydas*) in Nicaragua. *Biotropica* 13: 49–58.
- Murray, K., A. Bazzi, C. Carter, A. Ehlert, A. Ham's, M. Kopec, Richardson, J., and H. Sokol. 1997. Distribution and mobility of lead in soils at an outdoor shooting range. *Journal of Soil Contamination* 6: 79–93.
- Omrani, M., V. Ruban, G. Ruban, and K. Lamprea. 2017. Assessment of atmospheric trace metal deposition in urban environments using direct and indirect measurement methodology and contributions from wet and dry depositions. *Atmospheric Environment* 168: 101–111.
- Ostapczuk, P., M. Burow, K. May, C. Mohl, M. Froning, B. Süßenbach, E. Waidmann, and H. Emons. 1997. Mussels and algae as bioindicators for long-term tendencies of element pollution in marine ecosystems. *Chemosphere* 34: 2049–2058.
- Parmar, T. K., D. Rawtani, and Y. K. Agrawal. 2016. Bioindicators: the natural indicator of environmental pollution. *Frontiers in Life Science* 9: 10–118.
- Prado, P. and K. L. Heck Jr. 2011. Seagrass selection by omnivorous and herbivorous consumers: determining factors. *Marine Ecology Progress Series* 429: 45–55.
- Prange, J. A. and W. C. Dennison. 2000. Physiological Responses of Five Seagrass Species to Trace Metals. *Marine Pollution Bulletin* 41: 327–336.
- Roméo, M., M. Gnassia-Barelli, T. Juhel, and A. Meinesz. 1995. Memorization of Heavy Metals by Scales of the Seagrass *Posidonia Oceanica*, Collected in the NW Mediterranean. *Marine Ecology Progress Series* 120: 211–218.
- Ross, P. and M. E. DeLorenzo. 1997. Sediment contamination problems in the Caribbean islands: Research and regulation. *Environmental Toxicology and Chemistry* 16: 52–58.
- Sanz, M. A., V. M. Gabet, V., and J.-L. Gonzalez. 2021. Inputs of Total and Labile Dissolved Metals from Six Facilities Continuously Discharging Treated Wastewaters to the Marine Environment of Gran Canaria Island (Canary Islands, Spain). *International Journal of Environmental Research and Public Health* 18: 1–16.
- Savoca, M. S., N. A. Abreo, A. H. Arias, L. Baes,

- M. Baini, E. Bergami, S. Brander, M. Canals, C. A. Choy, I. Corsi, B. D. Witte, C. Domit, S. Dudas, E. M. Duncan, C. E. Fernández, M. C. Fossi, O. Garcés-Ordóñez, B. J. Godley, D. González-Paredes, V. G. Carman, B. M. Hamilton, B. D. Hardesty, S. H. Hong, S. Kahane-Rapport, L. M. Kashiwabara, M. B. Lacerda, G. Luna-Jorquera, C. Manano, S. E. Nelms, C. Panti, D. J. Pérez-Venegas, C. K. Pham, J. F. Provencher, S. Purca, H. Rashid, Y. Rodríguez, C. Sparks, C. Sun, M. Thiel, C. Tsangaris, and R. G. Santos. 2025. Monitoring plastic pollution using bioindicators: a global review and recommendations for marine environments. *Royal Society of Chemistry* 4: 10–32.
- Schlacher-Hoenlinger, M. A. and T. A. Schlacher. 1998. Accumulation, contamination, and seasonal variability of trace metals in the coastal zone—patterns in a seagrass meadow from the Mediterranean. *Marine Biology* 131: 401–410.
- Schneider, L., W. A. Maher, J. Potts, A. M. Taylor, G. E. Batley, F. Krikowa, A. Adamack, A. A. Chariton, and B. Gruber. 2018. Trophic transfer of metals in a seagrass food web: Bioaccumulation of essential and non-essential metals. *Marine Pollution Bulletin* 131: 468–480.
- Shahjahan, M., K. Taslima, M. S. Rahman, M. Al-Emran, S. I. Alam, and C. Faggio. 2022. Effects of heavy metals on fish physiology—A review. *Chemosphere* 300: 1–18.
- Shaw, K. R., G. H. Balazs, T. T. Jones, H. W. Lynch, J. Liu, G. P. Cobb, D. M. Klein, and J. M. Lynch. 2023. Green Sea Turtles (*Chelonia mydas*) Accumulate Heavy Metals Near a Former Skeet Shooting Range in Kailua, O’ahu, Hawai’i. *Environmental Toxicology and Chemistry* 42: 1109–1123.
- Siung-Chang, A. 1997. A review of marine pollution issues in the Caribbean. *Environmental Geochemistry and Health* 19: 45–55.
- Snijders, K. 2023. Urban Wastewater Management of Willemstad, Curaçao, and Environmental Implications. Available at: <https://repository.tudelft.nl/record/uuid:9112850b-9f17-4c4b-8f57-9db54c517e88>
- Sutherland, R. A. and F. M. G. Tack. 2002. Determination of Al, Cu, Fe, Mn, Pb and Zn in certified reference materials using the optimized BCR sequential extraction procedure. *Analytica Chimica Acta* 454: 249–257.
- Tripepi, M., I. J. R. van Veghel, A. D. Vreugdenhil, and E. Brunelli, E., 2025. First report of fibropapillomatosis and critical habitat use in green sea turtles in Curaçao. *The European Zoological Journal* 92: 925–934.
- Tupan, C. I. and R. Azriagishih. 2016. Accumulation and deposition of lead heavy metal in the tissues of roots, rhizomes and leaves of seagrass *Thalassia hemprichii* (Monocotyledoneae, Hydrocharitaceae). *Aquaculture, Aquarium, Conservation & Legislation-International Journal of the Bioflux Society* 9: 580–589.
- Unsworth, R. K. F. and L. C. Cullen-Unsworth. 2017. Seagrass Meadows. *Current Biology* 27: PR443–R445.
- USDA. 2021. Elemental Analysis Manual for Food and Related Products. Available at: <https://www.fda.gov/media/153993/download?attachment>
- Waait Institute. 2017. Marine Scientific Assessment: The State of Curaçao’s Coral Reefs. Available at: <https://openresearch.amsterdam/nl/page/76071/the-state-of-cura%C3%A7ao-s-coral-reefs>
- Wang, S., J. Wen, T.-Q. Guo, H.-J. Zhang, Y.-D. Song, G.-P. Fu, and T.-T. Wang. 2025. Assessment of heavy metal ecological risk in typical seagrass beds of the China Yellow Sea. *Aquatic Ecology* 59: 83–94.
- Wilkinson, A., E. Ariel, J. Van De Merwe, and J. Brodie. 2022. Trace element concentrations in forage seagrass species of *Chelonia mydas* along the Great Barrier Reef. *PLoS ONE* 17: e0269806.